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Jurassic magnetostratigraphy, 4. Early Callovian through Middle Oxfordian of the Krakow Uplands (Poland)

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ABSTRACT

A magnetic polarity pattern has been constructed for the Early Callovian through Middle Oxfordian stages of the Jurassic from several ammonite-rich magnetostratigraphy sections within the Krakow–Czestochowa–Wielun region of southern Poland. These overlapping sections encompass portions of every ammonite zone from the late-Early Callovian through late-Middle Oxfordian; however, the presence of several hiatuses and condensed intervals within the shallow-marine to pelagic sediments preclude reliable identification of the complete polarity pattern. The mean Callovian–Oxfordian pole from these sites is at 74.3°N, 200.3°E ($\alpha_{95} = 4.7°$).

The Callovian through Early Oxfordian is dominated by reversed polarity with a minimum of five normal-polarity zones. The early-Middle Oxfordian is predominantly of normal polarity, and the late-Middle Oxfordian is characterized by reversed polarity, with several relatively brief normal-polarity episodes. The Callovian and Oxfordian stages appear to average a minimum of two magnetic polarity reversals per million years. This reversal frequency is similar to the average Tertiary record, but is less than the reported spacing of Callovian and Oxfordian magnetic anomalies in the Pacific.

1. Introduction

The magnetic polarity patterns contained within sedimentary rocks provide an important tool for stratigraphic correlations. When calibrated to biostratigraphic events and zones, the resulting magnetic polarity time scale can be used to assign precise geological ages to the corresponding magnetic anomalies of oceanic crust and the equivalent magnetic polarity zones in strata poor in fossils. Such correlations of biostratigraphy to magnetostratigraphy have been accomplished in detail for the Tertiary [1,2], for the Cretaceous [3–12], and for the Kimmeridgian and Tithonian stages of the Late Jurassic [7–13]. The corresponding correlations to marine magnetic anomalies provide the nomenclature for polarity chrons of the magnetic polarity time scale for these time intervals.

The oldest well-documented magnetic anomaly, M25, has a corresponding polarity chron near the Oxfordian–Kimmeridgian stage boundary [13]. While closely spaced, low-amplitude magnetic anomalies older than M25 are reported from the Pacific [14–18], this Pacific pattern cannot be correlated with the broad low-amplitude anomalies of the oldest Atlantic crust [19,20]. Possible factors preventing correlation of these pre-M25 anomalies include topographic effects [21] and the difficulty of resolving brief polarity events in the slower-spreading Atlantic oceanic crust [22].

Callovian and Oxfordian sections have yielded conflicting magnetostratigraphy results. An extended interval of normal polarity with only one or two brief reversed-polarity zones is indicated by some studies in strata of this age span [23–25]. However, several other studies have observed mixed or predominantly reversed polarity [26–35]. Most of these early studies had either inadequate biostratigraphy, discontinuous stratigraphic sampling, or intervals of unreliable magnetizations, and the discrepancies cannot be resolved [36].

In order to construct a magnetic polarity time scale, it is essential to verify that the biostratigraphic–magnetostratigraphy correlations obtained in one section can be reproduced in other

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sections. Middle to Upper Oxfordian pelagic limestones of Aguilon in northern Spain have yielded the only reproducible magnetic polarity pattern, albeit from stratigraphic sections located within a kilometer of each other. The Aguilon magnetostratigraphy encompasses the upper *Gregoryceras transversarium* ammonite Zone (uppermost portion of the *Perisphinctes parandieri* Subzone) of late-Middle Oxfordian through the middle of the *Epipeltoceras bimammatum* Zone (*E. bimammatum* Subzone) of the middle-Late Oxfordian [36,37]. The general polarity pattern consists of three main polarity zones (reversed–normal–reversed) subdivided by shorter episodes. The net implied frequency of magnetic reversals is approximately four to six reversals per million years—a frequency similar to that of the Miocene–Pliocene stages of the Tertiary and to that reported from the Bathonian–Bajocian stages of the Middle Jurassic [38].

Callovian and Lower Oxfordian strata at Aguilon, and at various other localities in northern Spain and southern Portugal, failed to yield polarity patterns because the magnetizations were either unstable, too weak, or completely overprinted with secondary fields (M.B. Steiner and J.G. Ogg, unpublished data).

In this current study, several magnetostratigraphy sections were obtained in ammonite-rich marine sediments west of Krakow, Poland. These results provide a general picture of the magnetic polarity pattern for most of the ammonite zones of the Callovian to Middle Oxfordian stages.

2. Geologic setting and stratigraphy

The southern portion of the Krakow–Wielun Upland is a set of horst blocks uplifted during the Miocene. The stratigraphy and ammonites of the exposed Middle and Late Jurassic marine sediments have been the subject of numerous studies (e.g. [39–44]). A general summary of the Callovian and Oxfordian stratigraphy follows, while details of individual sections are presented on the magnetostratigraphy columns.

The late-Early Callovian through Late Oxfordian is represented by a depositional sequence as the sea transgressed, then withdrew, from the Krakow region. The uppermost Bathonian and basal ammonite zones of the Callovian are not documented paleontologically. Lower Callovian strata (*Sigaloceras calloviense* Zone) consist of sandstones and sandy crinoidal limestones containing bivalves, brachiopods, corals, borings and *Thallasinoid*-type burrows. The Middle Callovian, where present, is poorly represented by condensed sections of clayey limestone containing grains of iron oolite. Deepening of the sea continued through the Late Callovian, and very condensed layers of iron-rich stromatolites [43], Fe-Mn crusts and glauconitic nodular marls with phosphatic concretions were locally deposited. An irregular submarine topography during the Callovian is evident in lateral changes in facies and thickness. The basal Oxfordian generally consists of yellowish clayey limestone to calcareous claystone. A deep shelf environment during the Early and Middle Oxfordian is indicated by relatively thick sections of light gray sponge-rich bioclastic clayey limestone and light gray limestone. Thick sections of Upper Oxfordian limestone to clayey limestone are overlain by Upper Cretaceous deposits. Average rates of sedimentation increased greatly from the Middle Callovian to the Late Oxfordian.

The exposed sections are fairly flat-lying, with dips of 7° or less. Miocene thrusting of the Carpathian mountains to the south apparently caused simple horst uplifts in the Krakow region.

Ammonite zones within the condensed Callovian and basal Oxfordian strata are assigned primarily on the basis of ammonite assemblages, because the exact placements of first and last occurrences of marker taxa are not possible. Only a few sections have continuous sedimentation spanning multiple ammonite zones. Therefore, with rare exceptions, the magnetostratigraphy of the various sections yield an indication of the polarity characteristics, for each ammonite zone, but not a complete polarity pattern. In contrast, within the overlying upper-Lower and Middle Oxfordian strata in some sections, the first occurrences of marker taxa enabled assignment of subzonal boundaries (for example, at Zalas [44]).

Southern Poland was within the Submediterranean faunal province during most of the Callovian and Oxfordian. The regional ammonite zonations follow the "standard zonation" of the Submediterranean Province as established in France [45–47], but with minor modifications which vary among paleontologists and with date

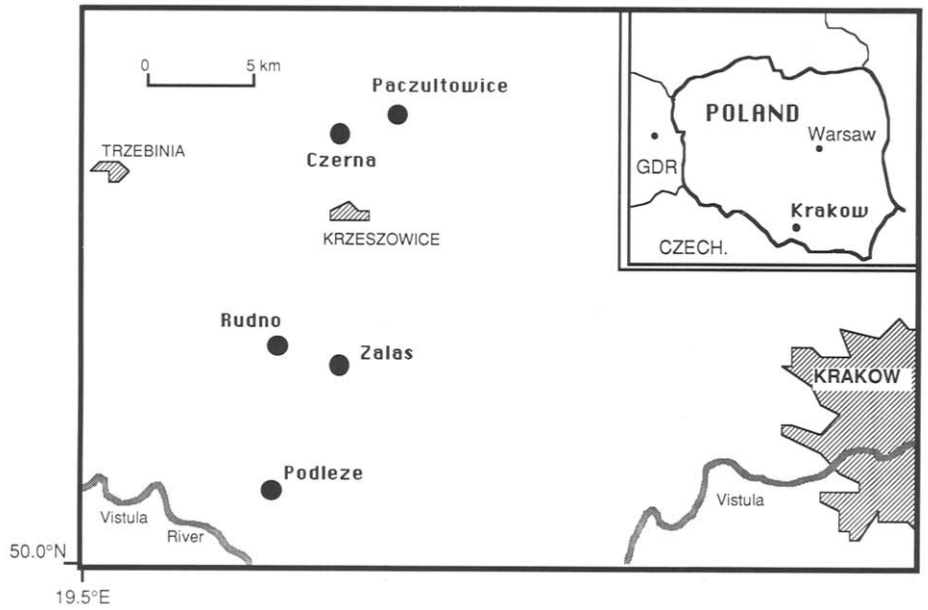


Fig. 1. Location of the magnetostratigraphy sections in the Krakow Uplands, southern Poland.

of publication. There is no fixed international standard zonation for these stages; however, correlation charts of these slight differences in nomenclature and subzonal divisions are available (e.g. [48]).

The five paleomagnetic localities are in the general vicinity of the city of Krzeszowice (50.1°N, 19.6°E; 25 km WNW of Krakow) (Fig. 1). The stratigraphic relations of all sections are summarized in the final composite diagram (Fig. 5).

(1) Paczoltowice is a village 5 km north-northeast of Krzeszowice; section "CMP" of Middle Callovian iron-oolite-rich limestone was excavated from a hillside farm road in the vicinity of "Stradlina." A total of 0.5 m was sampled.

(2) Czerna is 3 km west of Paczoltowice (or

Fig. 2. Vector plots of thermal demagnetization behavior of typical samples (all pilots except E and G). Declination and inclination vectors (solid and open dots, respectively) correspond to the horizontal component and to the actual dip of the magnetization vector; vectors have been corrected for tilt of strata.

A. Sample OP 22.6 m (Podleze; Middle Oxfordian, white limestone), characteristic direction (330–500°C) = 202.2° decl., −37.3° incl. (reversed polarity); one scale division = 2×10^{-6} A/m. Diagram enlarged, sharp drop from NRM (1.5×10^{-4} A/m; 262° decl., 83° incl.) to the 180°C step and viscous magnetic behavior above 550°C not shown.

B. Sample OCZO 1.72 m (Zalas; Middle Oxfordian, sponge-rich limestone), characteristic direction (240–500°C) = 16.6° decl., 42.4° incl. (normal polarity); one scale div. = 5×10^{-5} A/m. Diagram enlarged, continuation to NRM (5.7×10^{-4} A/m) indicated; viscous magnetic behavior above 500°C not shown.

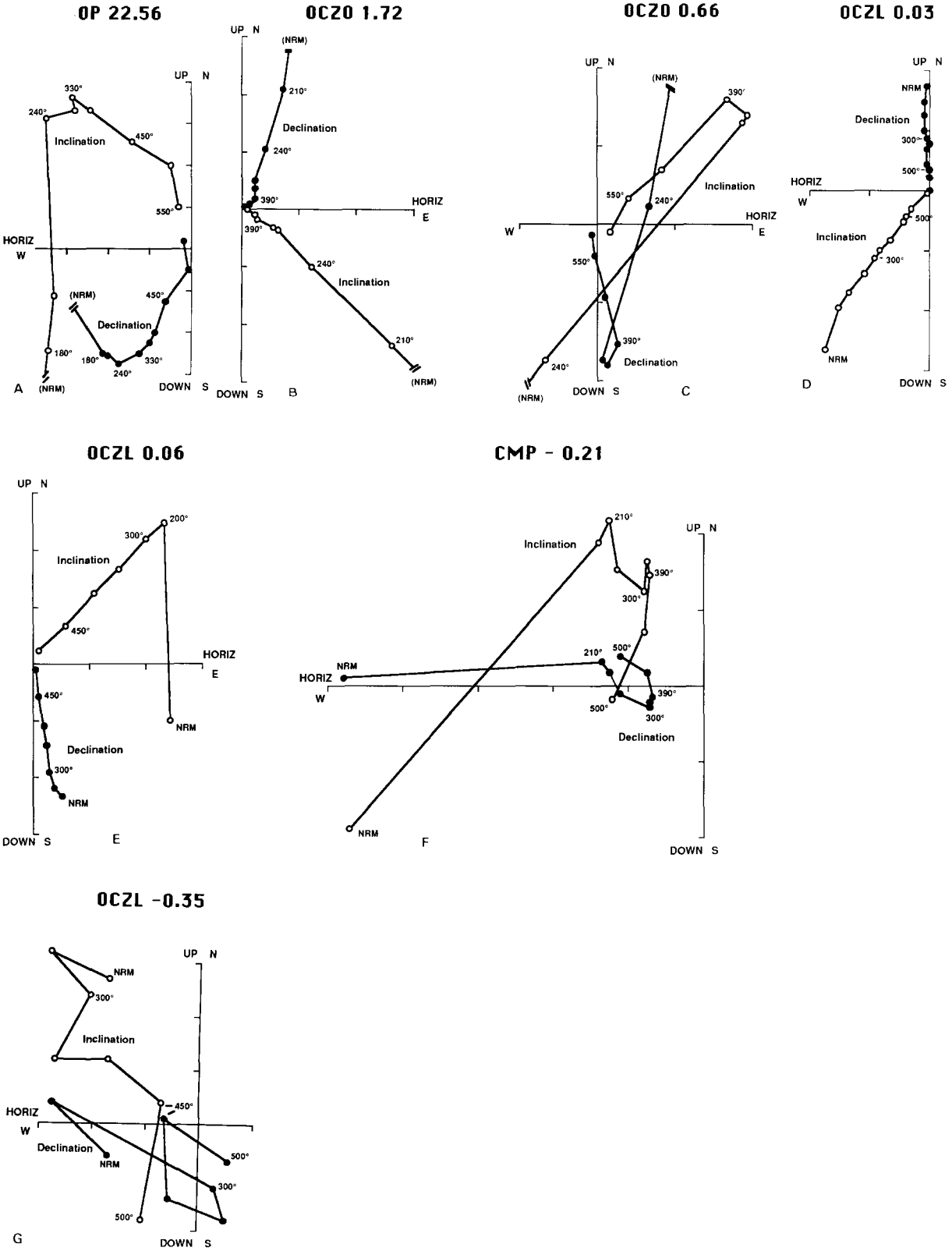
C. Sample OCZO 0.66 m (Zalas; Lower Oxfordian, marly limestone), characteristic direction (330–550°C) = 174.7° decl., −41.2° incl. (reversed polarity); one scale div. = 5×10^{-6} A/m. Diagram enlarged, continuation to NRM (2.6×10^{-3} A/m) indicated.

D. Sample OCZL 0.03 (Zalas; Upper Callovian, stromatolitic limestone), characteristic direction (300–600°C) = 356.2° decl., 52.8° incl. (normal polarity); one scale div. = 1×10^{-2} A/m.

E. Sample OCZL 0.06 m (Zalas; Upper Callovian, stromatolitic limestone), characteristic direction (300–500°C) = 172.8° decl., −46.6° incl. (reversed polarity); one scale div. = 1×10^{-4} A/m.

F. Sample CMP −0.21 m (Paczoltowice; Middle Callovian, oolitic limestone), characteristic direction (average of 300, 330 and 390°C) = 247.7° decl., −66.7° incl. (reversed polarity, "poor" quality); one scale div. = 2×10^{-4} A/m. Viscous magnetic behavior above 500°C not shown.

G. Sample OCZL (Zalas; Lower Callovian, sandy crinoidal limestone), characteristic direction (300–450°C) = 175.7° decl., −40.0° incl. (reversed polarity, "poor" quality); one scale div. = 1×10^{-4} A/m.


OP 22.56
OCZO 1.72
OCZO 0.66
OCZL 0.03
OCZL 0.06
CMP - 0.21
OCZL -0.35
UP N
DOWN S
HORIZ
W
E
Inclination
Declination
(NRM)
NRM
A
B
C
D
E
F
G

5 km NNW of Krzeszowice). Sections "CMCZ" and "CMCB" are on a hilltop 0.5 km east of the village and are of Middle Callovian iron-oolitic-rich limestone overlain by stromatolitic limestone of latest Callovian. It was not possible to accurately determine the amount of overlap between these two Czerna sections. A total of approximately 0.7 m was sampled.

(3) Zalas is 7 km south of Krzeszowice. Three sections were collected in the upper levels of a large active quarry 1 km east of the village. The stratigraphy and paleontology of the Zalas sections have been studied in detail (e.g. [44,49–51] and the paleomagnetism of an upper Callovian iron-rich stromatolite had previously been studied [32,33,52]. Sections "OCZL" and "OCZS" overlap in this Callovian interval, with "OCZL" extending into Lower Oxfordian clayey limestone; the composite section encompasses 1.3 m. Section "OCZO" is of Middle Oxfordian clayey and sponge-rich limestone (6 m).

(4) Rudna section "OCRD" is along a new divided-highway roadcut 6 km southwest of Krzeszowice. Lower Callovian sandy crinoidal limestone is overlain by Upper Callovian glauconitic marl and stromatolitic limestone; a total of 1.7 m was sampled. An additional 6-m section was collected in a Middle Oxfordian sponge-rich limestone bioherm exposed along a nearby highway overpass.

(5) Podleze sections are located on a hillside about 1.5 km southwest of Brodla or 12 km south-southwest of Krzeszowice. Sections #1 ("OCP") and #2 ("OCPT") are in a trench exposing Callovian strata similar to Rudna, overlain by basal Oxfordian stromatolitic limestone and oncolite-rich clayey limestone followed by lower-Middle Oxfordian greenish clayey limestone. The sampled intervals are 1.4 m and 0.4 m, respectively. Section #3 ("OP") is within an old quarry of upper-Middle Oxfordian white pelagic limestone and spans approximately 5 m.

3. Analytical procedures and magnetic properties

A total of 204 minicores of 2.5 cm diameter were drilled from the ten stratigraphic sections. Density of sampling depended upon the sedimentation rates, with an average of 10 to 20 samples collected in stratigraphic sequence from each ammonite zone in each section. Most beds had multiple samples. Paleomagnetic measurements and demagnetizations were performed in a steel-shielded room at the University of Wyoming, using a computer-linked, two-axis ScT cryogenic magnetometer. A pilot suite of the various lithologies from each section was treated by progressive thermal demagnetization in 12 steps up to 650°C; the results guided the progressive demagnetization of the remaining samples of each lithology.

Examples of typical magnetic behaviors for the main lithologies of each age are diagrammed in Fig. 2. The polarity of samples was generally apparent from the initial natural remanent magnetization (NRM) or upon heating to 200°C. Above 500°C, most lithologies displayed weak magnetizations or unstable magnetic directions. This general magnetic behavior suggests that the characteristic directions are carried primarily by magnetite, titanomagnetite and/or maghemite. A previous study of the rock magnetic properties of some of these same lithologies [34] concluded that goethite and hematite are important carriers of magnetization in the stromatolite intervals and that maghemite is a major magnetic phase in most other lithologies.

Most samples attained a direction that remained stable upon further demagnetization as the intensity of remanence decreased; hence, these displayed univectoral decay to the origin on the vector diagrams. Intensities of these characteristic magnetizations ranged from 10^{-5} to 10^{-3} A/m, with the NRM intensities generally an order of magnitude greater. Condensed intervals, such as the iron-rich stromatolite layers of latest Callovian, had the highest intensities of magnetization, with a hematite carrier of characteristic magnetization being apparent at higher thermal demagnetization steps.

Directions of characteristic magnetization and associated variances were computed for each sample by applying a three-dimensional least-squares line fit method or "principal component analysis" [53] which utilizes vector diagrams to identify linear removal of magnetic vectors. If the standard deviation of the removed vector was greater than 10°, or if fewer than three demagnetization steps were used in the computation, then the quality of these characteristic directions was considered to be "poor"; these samples are denoted by "P" in

the stratigraphic column of polarity interpretation. Results from samples having indeterminate or questionable polarity interpretations are denoted by "?" or "INT" in the stratigraphic columns and were omitted both from the polarity stratigraphy and from the computation of means directions for each site.

4. Mean directions

Mean directions of characteristic magnetization for normal-polarity and for reversed-polarity samples of each site were calculated using a method modified from Fisher [54]. The computational procedure gives half-weight to samples that have characteristic directions considered to be of "poor" quality (as explained above) and omits samples of questionable or indeterminate polarity. After the first iteration, any samples having directions falling outside two standard deviations from the mean are omitted from the final computation. Mean directions were also computed for each site as all samples combined and as the sum of the normal- and reversed-polarity directional vectors (Table 1). For sites having sections covering extended intervals of geological time, mean directions were also computed by geological age. Site means of characteristic directions of magnetization are displayed in Fig. 3.

Mean directions for normal-polarity vs. reversed-polarity samples at each site are antipodal to within 10° or to within the α_{95} for these mean directions. This "antipodal test" helps support the interpretation that the polarity zones represent magnetization set near the time of deposition. However, the mean intensity of magnetization of reversed-polarity samples is less than that of normal-polarity samples at some sites, which indicates that a secondary northward-directed overprint is present on some samples. For this reason, the *vector sum* of the normal-polarity and reversed-polarity directions, which incorporates the mean intensities of their magnetizations, is considered to be a more accurate representation of the characteristic mean direction for each site. Any secondary magnetic vector that affects equally samples of both polarity will be removed by such a vector addition method.

The upper Middle Oxfordian limestone of Section OP at Podleze has a mean direction that is much steeper dipping (inclination = 65°) than directions at other sections of Podleze or at the other sites. The bedding planes of this sponge-rich limestone have an apparent dip of toward the south, unlike the regional and local dip of formations toward the west. This is considered to be an artifact of sampling the side of a sponge bioherm; therefore, the bedding is representative of the paleo-slope, not paleo-horizontal. If the regional dip is applied, then the Middle Oxfordian direction of OP has an inclination of about 54°, which is consistent with the other sites.

An all-site mean direction was computed from the vector-sum means of the various localities (omitting the high-scatter results from Paczoltowice, and using the upper Callovian–lower Oxfordian subset from Zalas and the OCPT–OCP subset of Podleze). The mean direction of magnetization of the four sites has a declination towards present North (359.7°) with an inclination of 53.9° ($\alpha_{95} = 3.6°$, $K = 665$). This average inclination is consistent with a paleolatitude of about 35°N, which is about 15° south of the present positions of these sites. The mean pole from these four sites is located at 74.3°N, 200.3°E ($\alpha_{95} = 4.7°$, $K = 379$) (Table 2).

The mean directions of Callovian through Middle Oxfordian sections or of geological age-subdivisions within these sections suggests that these sites experienced a slight amount of northward drift between the Callovian and Oxfordian with no significant accompanying rotations. However, the indicated amount of northward drift may not be statistically significant.

An earlier paleomagnetic study of the Upper Callovian iron-rich stromatolite layer at Zalas yielded normal polarity and a paleomagnetic direction of 2.8° declination, 40.0° inclination ($\alpha_{95} = 2.6°$) [32,33,52]; our results are similar. This earlier study also indicated that at Podleze, the stromatolitic layer of latest Callovian has reversed polarity overlain by normal polarity with a mean direction of about 340° declination, 57° inclination, and that a block of Middle Oxfordian limestone yielded normal polarity with a direction of 347°, 59° ($\pm 3°$). These results also agree with our present study.

Kadzialko-Hofmokl and Kruczyk [34] have computed a mean pole for Middle to Late Jurassic "stable Europe" from earlier paleomagnetic stud-

TABLE 1

Mean directions of characteristic magnetization (after structural correction)

Site, age, polarity	N/N_t	Mean Direction				Mean intensity (10^{-4} A/m)	VGP			
		decl. (°)	incl. (°)	α_{95} (°)	K		lat. (°N)	long. (°E)	(dp/dm) (°)	lat. (°N)
Paczultowice										
M. Callovian (CMP)										
N	2.5/3	342.8	59.5	44.1	14	4.6	74.5	257.1	(49.7/66.2)	40.3
R	6/8	41.6	50.1	26.8	7	3.0	53.6	125.6	(23.9/35.8)	30.8
Combined	8.5/11	27.2	55.8	21.9	7	3.4	66.1	134.4	(22.4/31.3)	36.3
N + R	8.5/11	9.4	59.3			3.6	78.0	162.6		40.1
Czerna										
M.-u. Callovian (CMCZ, CMCB)										
N	4.5/6	7.1	54.0	32.2	8	3.4	73.6	178.4	(31.5/45.0)	34.5
R	5.5/7	351.9	55.1	11.9	37	2.1	74.4	224.7	(12.0/16.9)	35.6
Combined	10/13	358.6	54.9	13.0	15	2.6	75.3	204.3	(13.0/18.4)	35.4
N + R	10/13	1.4	54.6			2.8	75.0	195.2		35.2
Zalas										
U. Callovian – m. Oxfordian (all sections)										
N	13/16	8.2	48.4	6.1	48	3.5	68.4	179.9	(5.2/8.0)	29.4
R	15/20	10.9	43.2	12.6	10	1.8	63.6	183.1	(9.7/15.6)	25.1
Combined	28/36	9.7	45.7	7.0	16	2.4	65.9	178.2	(5.6/8.9)	27.1
N + R	28/36	9.2	46.6			2.6	66.7	178.7		27.9
U. Callovian – l. Oxfordian (OCZL, OCZS)										
N	6.5/7	1.7	54.2	7.4	74	11.3	74.6	194.4	(7.3/10.4)	34.8
R	8/11	11.0	54.3	9.9	32	3.8	72.8	167.7	(9.8/13.9)	34.8
Combined	14.5/18	6.8	54.4	6.0	44	6.2	74.0	179.0	(5.9/8.4)	34.9
N + R	14.5/18	4.0	54.3			7.5	74.5	187.3		34.9
L.-m. Oxfordian (OCZO)										
N	8/9	20.9	39.3	12.5	21	0.93	57.7	161.5	(8.9/14.9)	22.2
R	6.5/9	7.0	27.6	22.7	9	0.74	54.1	188.0	(13.5/24.7)	14.6
Combined	14.5/18	14.4	34.4	11.6	12	0.84	56.8	174.2	(7.6/13.3)	18.9
N + R	14.5/18	14.3	34.3			0.83	56.7	174.4		18.8
Rudno										
L. Callovian – m. Oxfordian (OCRD)										
N	6.5/9	358.6	56.5	14.5	20	8.4	76.9	204.5	(15.1/20.9)	37.0
R	13.5/17	8.1	45.9	9.2	20	2.3	66.4	181.3	(7.5/11.7)	27.3
Combined	20/26	5.5	49.4	7.6	19	3.5	69.8	185.8	(6.7/10.1)	30.3
N + R	20/26	1.1	54.3			5.4	74.7	196.3		34.8
L. Callovian (lower OCRD)										
N	2/4	343.4	43.4	56.7	3	3.5	62.1	233.1	(43.9/70.5)	25.3
R	5.5/8	10.3	47.1	15.2	23	4.9	66.8	176.0	(12.7/19.6)	28.2
Combined	7.5/12	3.8	46.8	19.0	10	4.5	67.7	190.8	(15.7/24.4)	28.0
M. Oxfordian (upper OCRD)										
N	5/5	357.2	58.0	10.6	53	10.6	78.4	210.5	(11.5/15.6)	38.6
R	8/9	6.6	45.1	13.8	17	1.4	66.0	184.9	(11.0/17.4)	26.6
Combined	13/15	3.6	50.2	9.3	21	3.0	70.7	190.4	(8.4/12.4)	31.0
N + R	13/15	358.6	56.5			5.9	76.9	204.5		37.1

TABLE 1 (continued)

Site, age, polarity	N/N_t	Mean Direction				Mean intensity (10^{-4} A/m)	VGP			
		decl. (°)	incl. (°)	α_{95} (°)	K		lat. (°N)	long. (°E)	(dp/dm) (°)	lat. (°N)
Podleze										
U. Callovian – m. Oxfordian (all sections)										
N	17/26	347.8	66.2	6.1	35	0.98	81.9	283.7	(8.2/10.0)	48.6
R	25/34	5.0	60.2	4.8	37	0.41	80.4	176.6	(5.6/7.3)	41.1
Combined	42/60	358.9	62.9	3.9	33	0.58	84.2	207.6	(4.8/6.1)	44.3
N + R	42/60	353.6	64.7			0.69	84.5	252.6		46.6
U. Callovian – m. Oxfordian (OCP, OCPT)										
N	8.5/15	347.6	65.1	8.9	37	2.7	81.3	274.4	(11.6/14.4)	47.1
R	4/7	355.8	38.2	36.8	7	2.5	61.2	207.8	(25.7/43.5)	21.4
Combined	12.5/22	351.3	57.1	12.4	13	2.6	76.1	229.7	(13.1/18.0)	37.7
N + R	12.5/22	352.8	52.0			2.5	71.8	219.2		32.7
M. Oxfordian (OP)										
N	8.5/11	347.9	67.3	9.8	31	0.36	82.3	294.6	(15.5/16.2)	50.1
R	21.5/27	5.0	62.5	4.4	52	0.36	82.8	169.4	(5.4/6.9)	43.7
Combined	30/38	0.8	64.0	4.1	41	0.36	85.6	192.2	(5.3/6.6)	45.7
N + R	30/38	357.2	65.1			0.36	86.6	232.7		47.2

N/N_t = number of samples after weighting/total samples (characteristic directions of poor stability are given half-weight). Questionable and indeterminant polarity samples are omitted. α_{95} = radius of circle of 95% confidence about mean direction. K = dispersion parameter. Mean intensity = logarithmic mean of the vectors incorporated in computation of characteristic directions. VGP = virtual geomagnetic pole (dp/dm = oval of 95% confidence). N,R = samples of normal and of reversed polarity, respectively. Combined = all samples combined. N + R = sum of mean normal and reversed direction vectors.

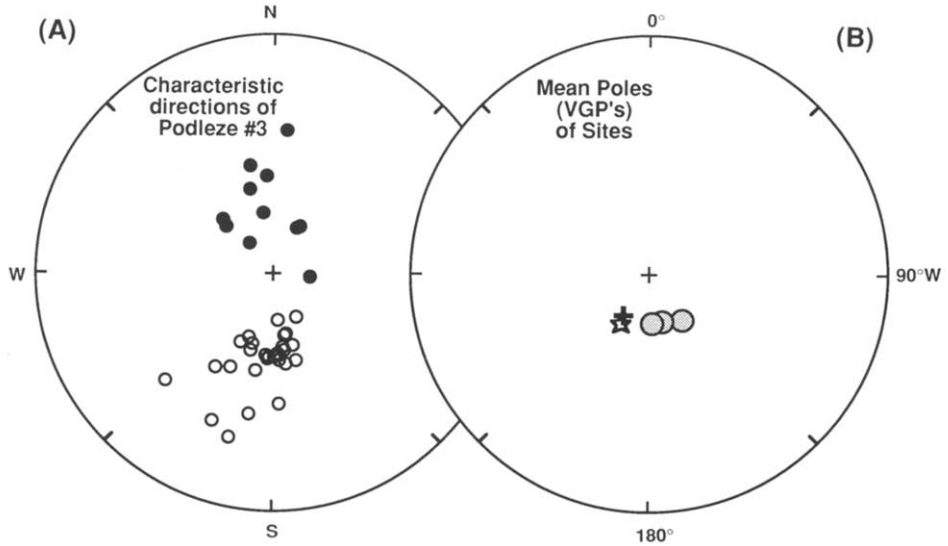


Fig. 3. A. Characteristic directions of magnetization from Podleze #3 ("OP"); equal-area plot, solid circles are positive inclination, open circles are negative inclinations. Samples with uncertain polarity are omitted. B. Mean poles (VGP's) from the four Krakow Upland sites (solid circles; Rudno and Czerna VGP's are nearly identical) of this study compared to the "mean Krakow Upland" and to the "stable Europe" Jurassic poles (star and cross, respectively) of Table 2.

TABLE 2

Mean direction and pole positions, Callovian-Oxfordian, Krakow Uplands

	"Sites"	Decl. (°E)	Incl. (°)	A_{95} (°)	K	Pole (°E, °N)
Present study	4	359.7	53.9	3.6	665	200.3, 74.3
Krakow Upland [34]	8	16.8	56.7	7.3	58	150.4, 72.3
"Stable Europe" [34]	12	15.7	59.0	5.0	75	147.1, 74.7

"Sites" = number of localities (present study, "Stable Europe") or number of exposures (Krakow Upland study consists of 6 localities). Decl., Incl. = declination and inclination of mean direction. A_{95} = radius of 95% confidence circle. K = dispersion parameter.

ies [25,55,56] (Table 2). The predicted mean direction for the Krakow Uplands is very close to their observed mean direction as computed from seven Callovian and one middle-Oxfordian sites. However, they omitted eight sites, including three at Podleze, which deviated significantly from the predicted pole. Our Podleze results, after omitting the sponge-bioherm samples, do not show significant divergence, possibly due to the averaging of more beds and lithologies.

Our present study yields a mean declination slightly clockwise of this computed direction of "stable Europe" [34] and a mean inclination slightly shallower. Such differences may merely indicate that this "stable Europe" pole, which averages a 50 m.y. interval of time, is not characteristic of the Callovian–Oxfordian, and that some minor plate motion of Europe occurred during the Middle through Late Jurassic.

5. Magnetic stratigraphy

The characteristic directions of magnetization, polarity interpretation, lithostratigraphy, and ammonite zonation for the ten sections are plotted stratigraphically in Fig. 4.

To facilitate visual comparison and correlation of the magnetic polarity patterns among sections, it was necessary to partially compensate for the extreme variation in sedimentation rates. Therefore, polarity patterns for each ammonite zone of each section (and for the composite polarity column from Aguilon, Spain) are plotted using a standard biostratigraphic scale in Fig. 5. This chart of polarity vs. age also dramatically illustrates for each section the durations of hiatuses and those intervals in which only a portion of an ammonite zone is presumed to be present. The main features of the magnetic polarity pattern for each ammonite zone within the different sections are compiled as a "composite" scale to give a very general outline of the polarity pattern for the Callovian and Oxfordian stages.

Although the upper-Early Callovian (*Macrocephalites* (*Dolikephalites*) *gracilis* or *Sigaloceras calloviense* Zone) sandy crinoidal limestones were sampled in three sections (Zalas #1, Rudno, and Podleze #1), many of these samples yielded data of "poor" quality. This interval is characterized by reversed polarity, with perhaps a brief normal-polarity zone.

Polarity columns from the two sections of Middle Callovian (*Kosmoceras* (*Zugokosmoceras*) *jason* and *Erymnoceras coronatum* Zones) iron-oolite-rich clayey limestone of Paczoltowice and Czerna do not agree in the relative proportions of normal- and reversed-polarity zones. This inconsistency is perhaps due to a difference in age that resulted from the inability to differentiate the ammonite zones. A minimum of two reversed-polarity and two normal-polarity zones is suggested.

The lower-Late Callovian (*Peltoceras athleta* Zone) is entirely of reversed polarity in the clayey limestone and calcareous claystone of Czerna and Rudno; however, the sample density in each section is inadequate to detect short polarity zones. As previously documented in detail [32,34], the iron-oxide-rich stromatolites of the latest Callovian (*Quenstedtoceras* (*Lamberticeras*) *lamberti* Zone) have a reversed-polarity zone overlain by a normal-polarity zone (Podleze #1). The uppermost portion of this *Q. lamberti* Zone in oncolite-rich clayey limestone is of reversed polarity. Other sections within the condensed stromatolite facies of the *Q. lamberti* Zone may have preserved only a portion of this pattern.

The lowest Oxfordian (*Quenstedtoceras* (*Quenstedtoceras*) *mariae* Zone) yielded reversed polarity at Zalas #1 and normal polarity at Podleze

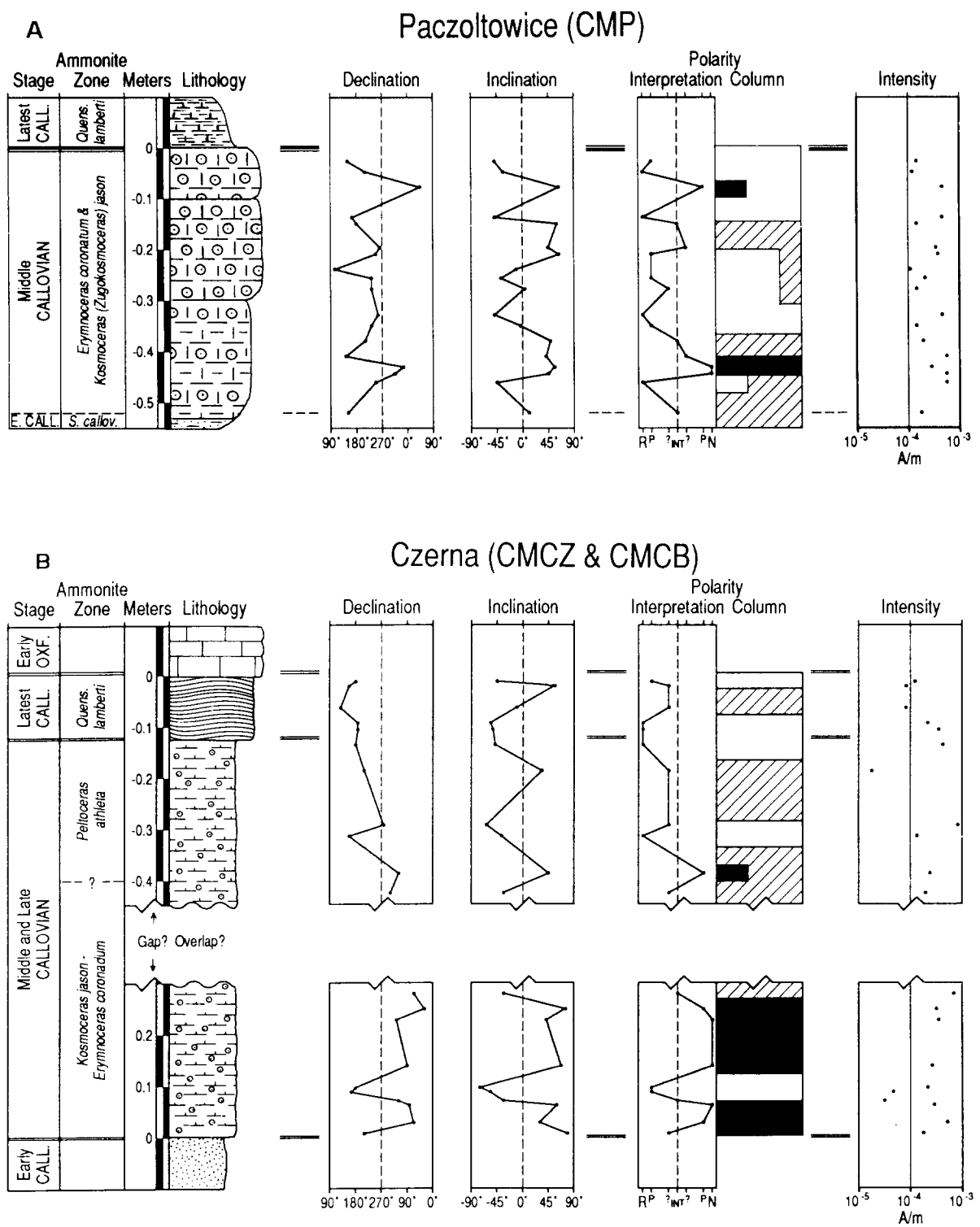
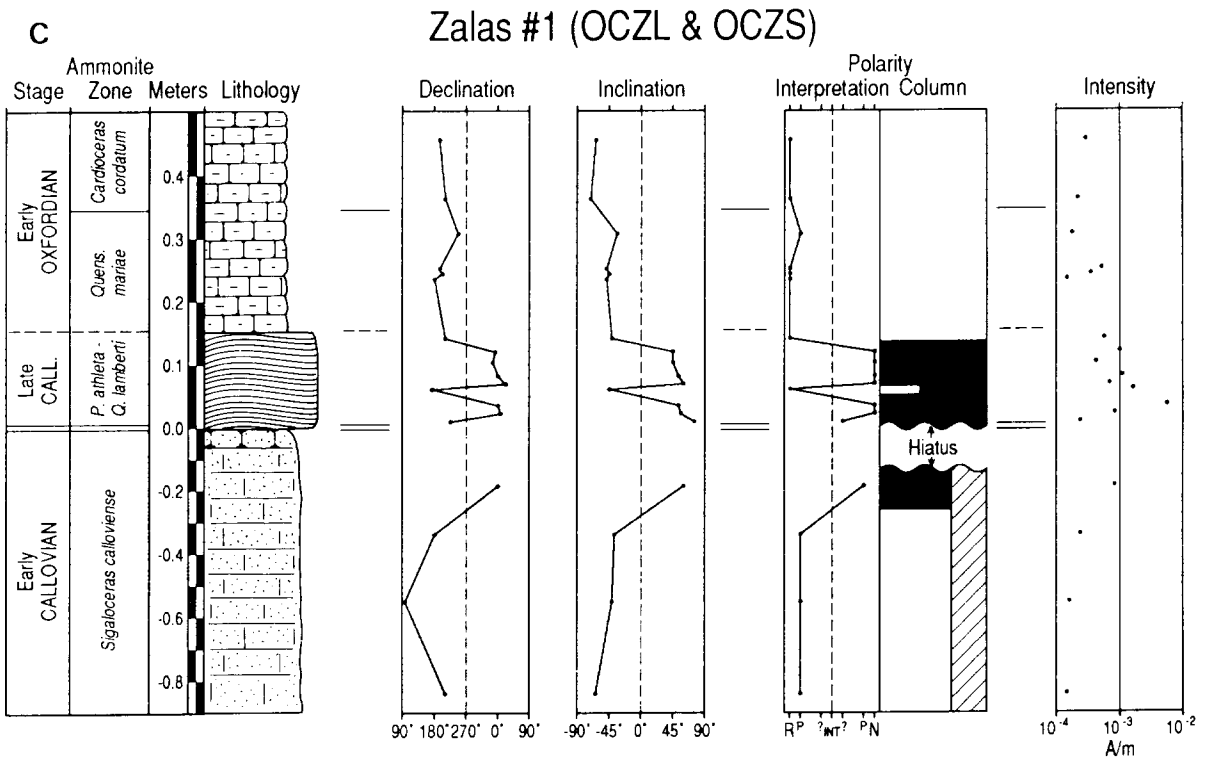


Fig. 4. Magnetostratigraphy sections from the Krakow Uplands with biostratigraphy, lithostratigraphy, characteristic directions of magnetization (declination, inclination, and intensity), and polarity interpretation (with an index of the reliability of the characteristic direction. In the polarity column, black is normal polarity.



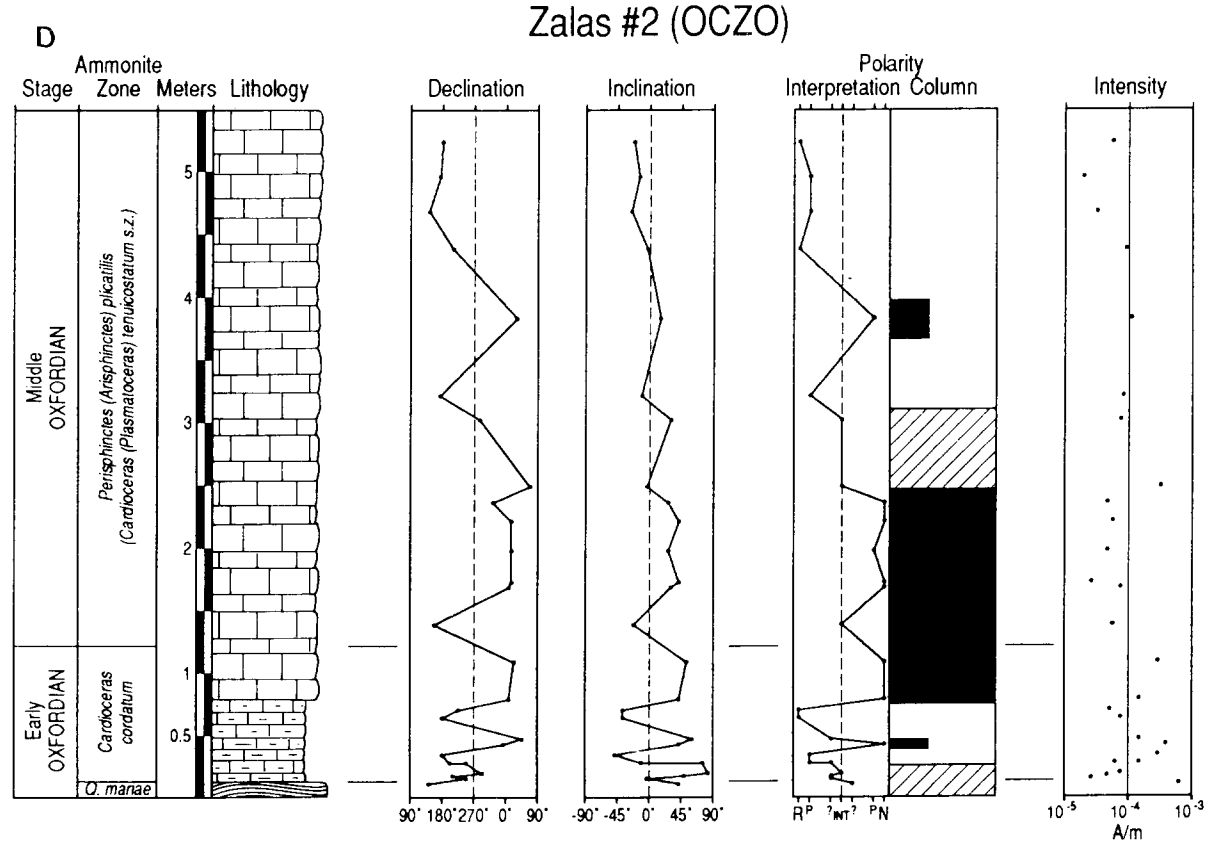


Fig. 4 (continued).

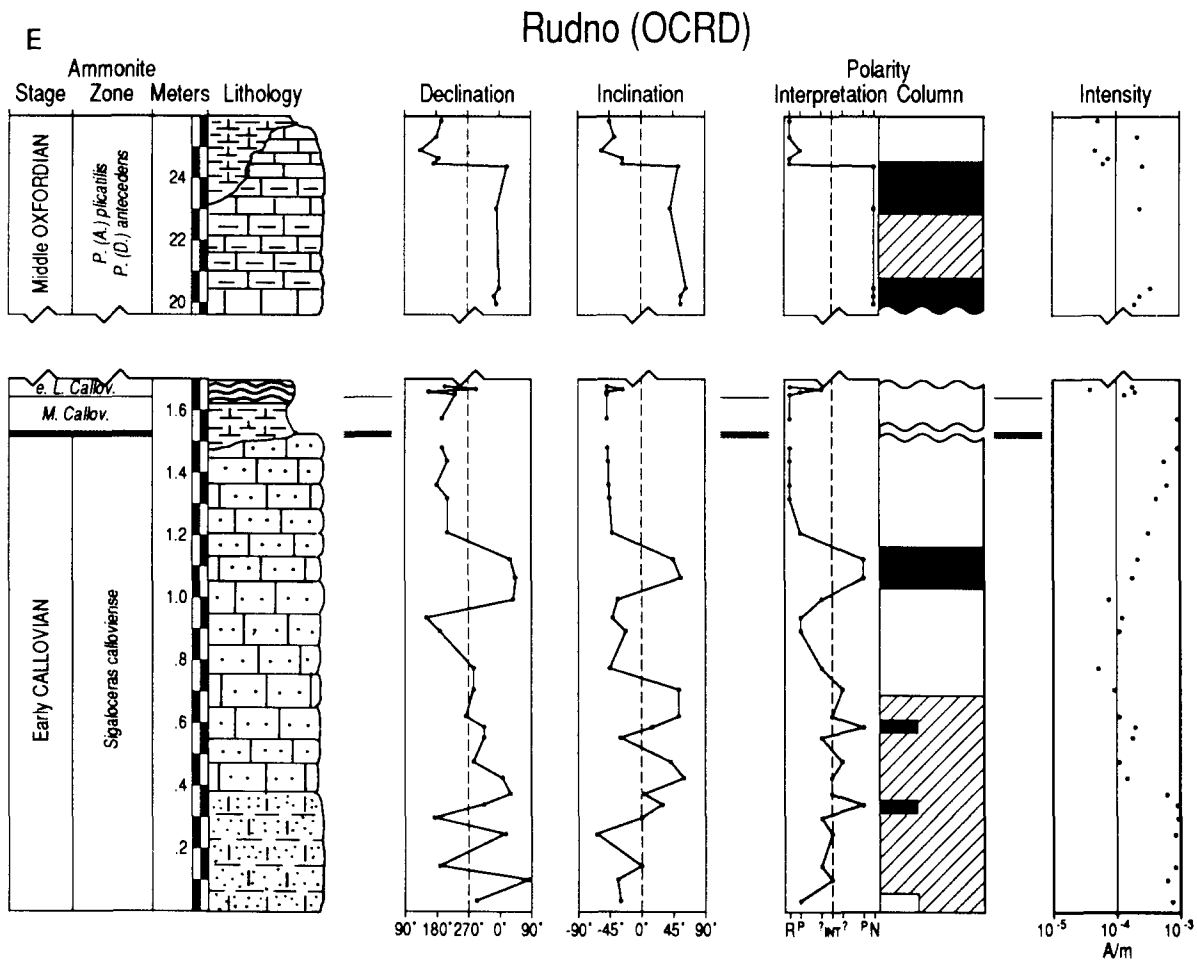


Fig. 4 (continued).

#2; however, the sampling density at each section is inadequate to compile a polarity pattern.

These Callovian through earliest Oxfordian results are not from complete or overlapping stratigraphic sections; therefore, the polarity columns do not provide the documentation and verification needed to construct a complete composite polarity pattern. A "possible" polarity sequence is shown that assumes all the magnetostratigraphy sections are accurate representations of the magnetic polarity history. The hatchured pattern indicates that this polarity sequence is not an accurate time scale.

In contrast to the Callovian, the sections of Early and Middle Oxfordian have more rapid and continuous sedimentation and provide better recorders of characteristic magnetizations.

The upper-Early Oxfordian (*Cardioceras cordatum* Zone) is of reversed polarity with an overlying normal-polarity zone extending across the Early/Middle Oxfordian boundary. The lower-Middle Oxfordian (*Perisphinctes* (*Arisphinctes*) *plicatilis* Zone) is dominated by normal polarity, especially within the upper ammonite subzone (*Perisphinctes* (*D.*) *antecedens* Subzone), as verified by three sections.

The upper-Middle Oxfordian (*Gregoryceras transversarium* Zone) portion of the Podleze #3 section may overlap with the lower portion of the Aguilon (Spain) magnetostratigraphy [36]. Both regions indicate that this ammonite zone is characterized by reversed polarity with brief normal-polarity intervals.

6. Conclusions

Several ammonite-rich sections within the Krakow Uplands region of southern Poland have

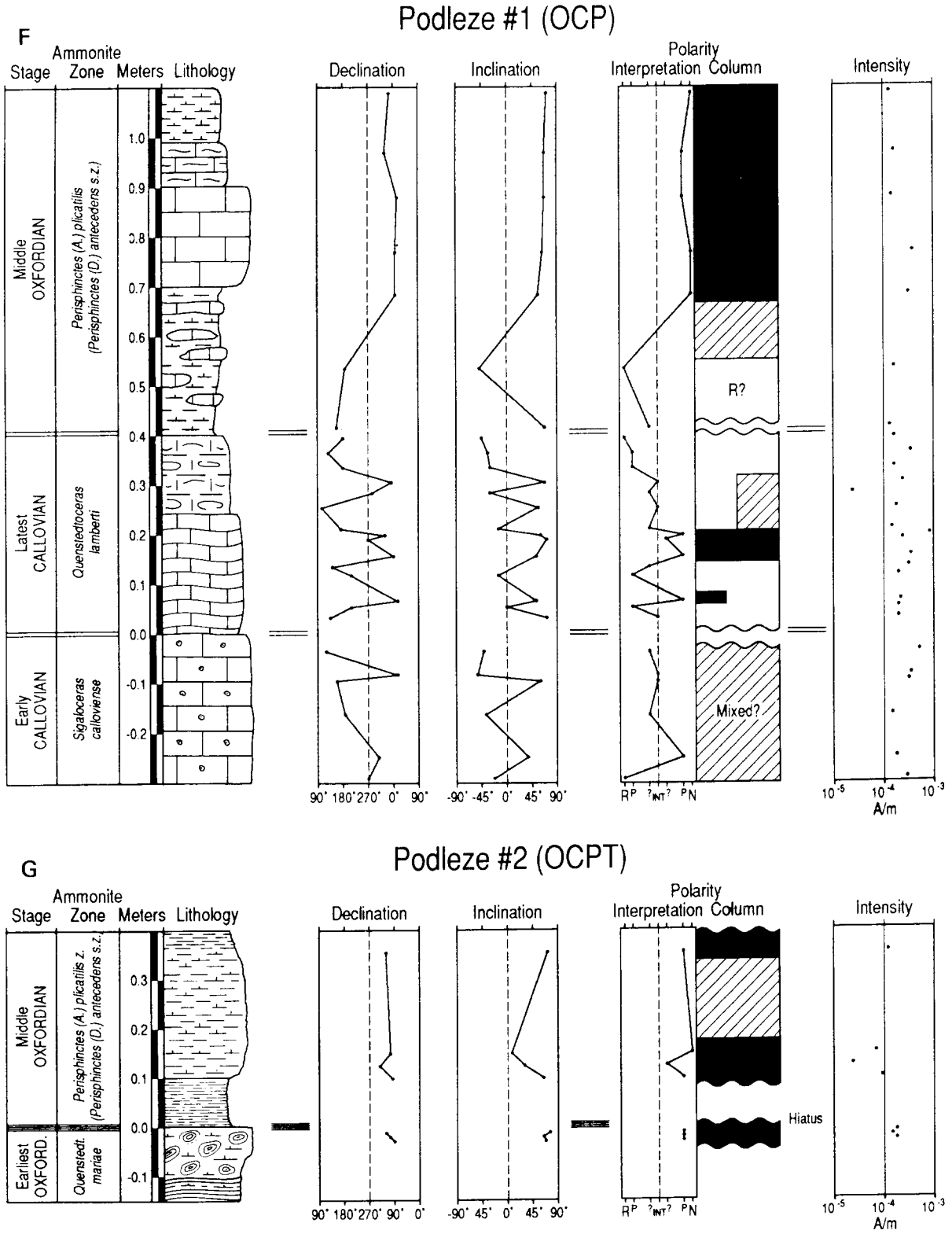


Fig. 4 (continued).

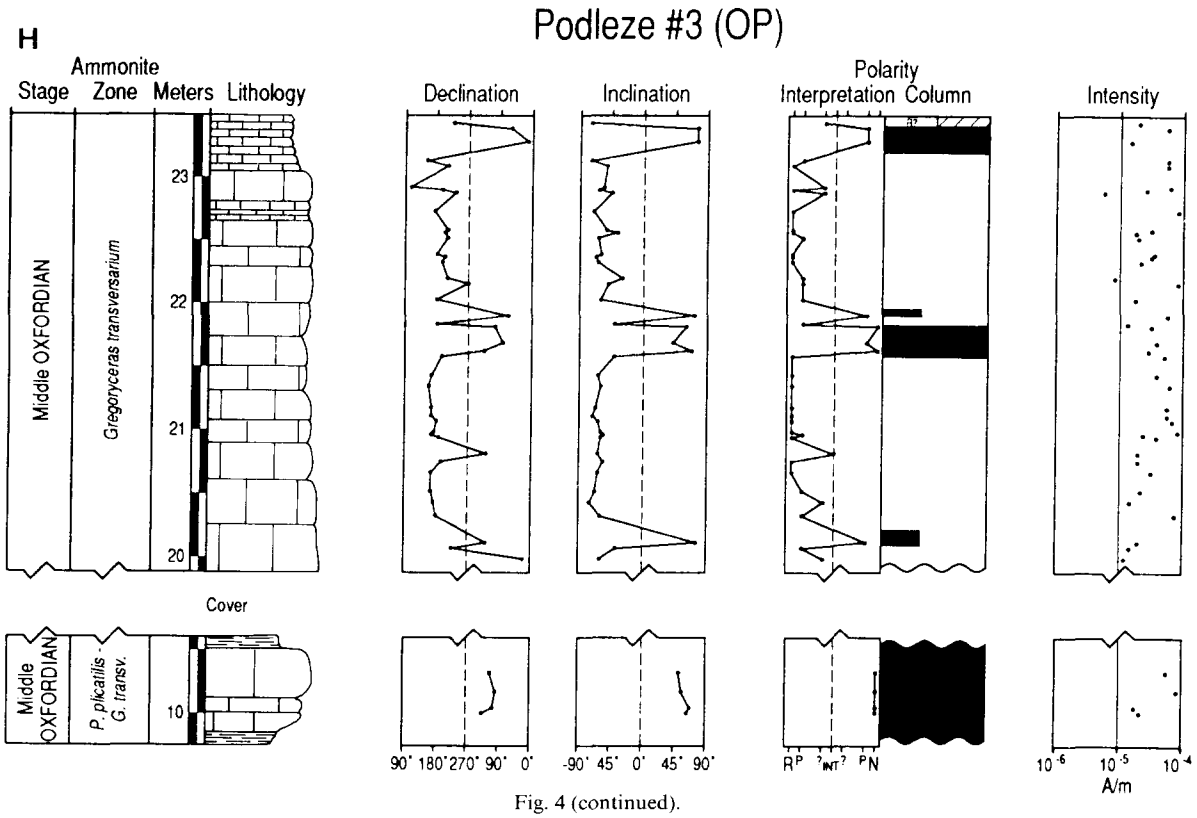


Fig. 4 (continued).

yielded magnetostratigraphy results. These overlapping sections encompass portions of every ammonite zone from the late-Early Callovian through late-Middle Oxfordian; however, the presence of several hiatuses and condensed intervals within the shallow-marine to pelagic sediments preclude reliable identification of the complete polarity pattern. Normal and reversed polarity is evident at NRM or upon heating to 200°C, and stable characteristic directions are obtained through 500°C.

The mean characteristic directions for normal and reversed polarity from each site are antipodal and are near the directions predicted from other studies of "stable Europe" Jurassic. The mean Callovian–Oxfordian pole from these sites is at 74.3°N, 200.3°E ($\alpha_{95} = 4.7°$).

The Callovian through Early Oxfordian is dominated by reversed polarity with a minimum of five normal-polarity zones. The early-Middle Oxfordian is predominantly of normal polarity, and the late-Middle Oxfordian is characterized by reversed polarity with several relatively brief normal-polarity episodes. The Callovian and Oxfordian stages appear to average about two magnetic polarity reversals per million years. This reversal frequency is similar to the average Tertiary record.

Pacific oceanic crust of presumed Oxfordian and Callovian age has yielded a set of 24 magnetic anomalies, "M26" to "M38", which formed within about an 8 m.y. interval of time and are considered to represent polarity episodes [15–18]. The Pacific pattern displays no distinctive extended intervals of constant polarity and the average frequency is predicted as three reversed-polarity zones per million years. This is about twice the frequency recorded in the Polish magnetostratigraphy sections. Therefore, either the Pacific had a slower spreading rate and a correspondingly longer time span is represented by this anomaly pattern, or else the condensed Polish sections preserved only about half of the actual number of magnetic polarity chrons within the Callovian through Middle Oxfordian. The results from these Poland sec-

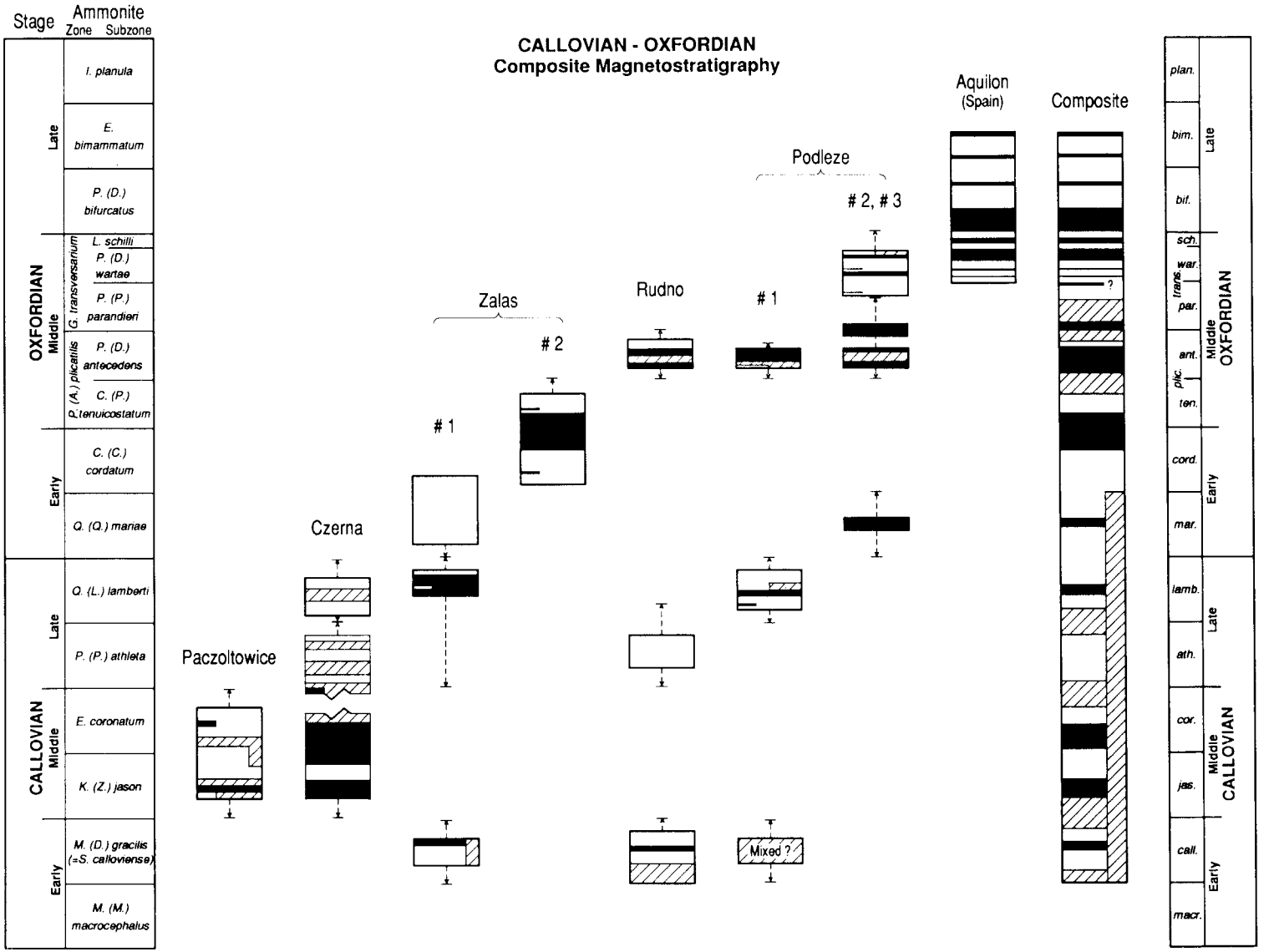


Fig. 5. Summary of all magnetostratigraphy sections from the Krakow Uplands and the composite polarity column from Aguilon [36], plotted with respect to ammonite zones. Black is normal polarity, white is reversed polarity; hatchured is uncertain results or gaps in sampling. Maximum uncertainty of the biostratigraphic age assignments is indicated by error bars. The portion of the polarity column drawn falls within the indicated ammonite zone, but it may not span all of that ammonite zone, due to hiatuses in sedimentation. The composite polarity pattern (on the left side) indicates the relative abundance of normal vs. reversed polarity and some of the major polarity features observed for each ammonite zone (hatchured shading indicates uncertain results). This composite polarity sequence is not to be considered as complete or as an accurate magnetic polarity time scale.

tions do indicate that these low-amplitude magnetic anomalies of the Pacific are not entirely due to intensity fluctuations of the Jurassic magnetic field, but record actual magnetic reversals. The main features of the composite polarity pattern derived from these Polish sections will require verification in thicker and more continuous sections having similar precise ammonite zonations.

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